

Jacobi Robotics Manipulation Take-Home Task

Analytic Inverse Kinematics

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1 Introduction

The robot is composed of a linear axis and a 6-Degree of Freedom (DoF) ABB robotic arm. The variables for the positions of the joints are denoted as $\mathbf{q} = (q_1, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$ with q_1 defining the scalar position of the prismatic joint of the linear axis, and θ_i for $i \in [1, 6]$ being the angular positions of the revolute joints of the 6-DoF robotic arm.

The frames attached to each link are firstly defined with subscripts denoted as follows:

- b : base frame as global reference frame attached to the ground;
- 0 : a reference frame for modeling the robot manipulator, attached to the linear axis base, which has a fixed transformation relative to the base frame;
- L : linear axis frame attached to the center of the prismatic joint of the linear axis;
- LF : linear axis' flange frame, on which the ABB robot arm is attached;
- $1 - 6$: robotic arm's frame located at the center of each revolute joint i (with angle being θ_i);
- EE : end-effector frame attached on the last link of the ABB robotic arm.

The analytic inverse kinematics (IK) problem is to find the set of joint configurations $Q = \{\mathbf{q}^i\}$ where $\mathbf{q}^i$ satisfies

$${}^bT_{EE} = {}^bT_{EE}(q_1, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6) = {}^bT_{EE}^d \quad (1)$$

where ${}^bT_{EE}^d$ is the transformation matrix corresponding to the desired end-effector pose, which can be represented by

$${}^bT_{EE}^d = \begin{bmatrix} t_{11} & t_{12} & t_{13} & p_x \\ t_{21} & t_{22} & t_{23} & p_y \\ t_{31} & t_{32} & t_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where p_x, p_y, p_z define the desired 3-dimensional position of the end-effector, and t_{ij} for $i \in [1, 3]$ and $j \in [1, 3]$ are the elements of rotation matrix representing the desired orientation of the end-effector.

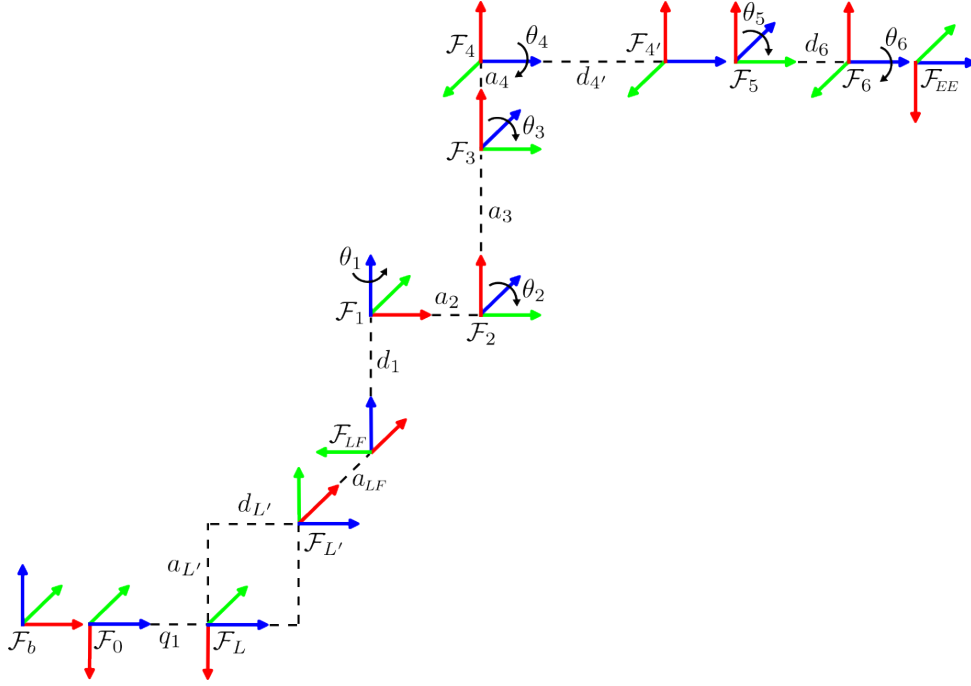


Figure 1: Definition of frames for Jacobi Robot composed of a linear axis and 6-DoF robotic arm. x , y , z axes of the frames are rendered in red, green and blue, respectively.

2 Forward Kinematics

In order to solve the joint configurations from a given end-effector pose, forward kinematics of the robot is at first derived, which aims to compute the end-effector pose from a given joint configuration $\mathbf{q}$. To achieve this, the modified Denavit-Hartenberg (MDH) method is adopted. The definition of frames are shown in Fig. 1. It should be noted that two additional frames are introduced, i.e., L' and $4'$ which are respectively fixed to linear axis and 4-th joint of the robotic arm, for the purpose of modeling the robot consistently with MDH convention.

The MDH table is defined in accordance with frame definition and as follows:

$\mathbf{Tf}$	α	a	θ	d
$0 \rightarrow L$	0	0	0	q_1
$L \rightarrow L'$	0	$a_{L'}$	$\frac{\pi}{2}$	$d_{L'}$
$L' \rightarrow LF$	$-\frac{\pi}{2}$	a_{LF}	0	0
$LF \rightarrow 1$	0	0	$\theta_1 - \frac{\pi}{2}$	d_1
$1 \rightarrow 2$	$-\frac{\pi}{2}$	a_2	$\theta_2 - \frac{\pi}{2}$	0
$2 \rightarrow 3$	0	a_3	θ_3	0
$3 \rightarrow 4$	$-\frac{\pi}{2}$	a_4	θ_4	0
$4 \rightarrow 4'$	0	0	0	$d_{4'}$
$4' \rightarrow 5$	$\frac{\pi}{2}$	0	θ_5	0
$5 \rightarrow 6$	$-\frac{\pi}{2}$	0	θ_6	d_6
$6 \rightarrow EE$	0	0	π	0

Table 1: Denavit-Hartenberg Parameters

where values for constants are provided as: $a_{L'} = -0.55$, $d_{L'} = 0.6$, $a_{LF} = 0.43$, $d_1 = 0.78$, $a_2 = 0.32$, $a_3 = 1.28$, $a_4 = 0.2$, $d_{4'} = 1.5925$, $d_6 = 0.2$ (in meters). Based on the

calculation of each individual transformation matrix using DH method and parameters, the end-effector pose in the frame 0 can be derived, which now takes the form of

$${}^0T_{EE} = {}^0T_L(q_1) \cdot {}^LT_{L'} \cdot {}^{L'}T_{LF} \cdot {}^{LF}T_1(\theta_1) \cdot {}^1T_2(\theta_2) \cdot {}^2T_3(\theta_3) \cdot {}^3T_4(\theta_4) \cdot {}^4T_{4'} \cdot {}^{4'}T_5(\theta_5) \cdot {}^5T_6(\theta_6) \cdot {}^6T_{EE} \quad (3)$$

with ${}^LT_{L'}$, ${}^{L'}T_{LF}$, ${}^4T_{4'}$ and ${}^6T_{EE}$ being fixed transformations. Additionally, the fixed transformation from frame 0 to base frame is given by

$${}^bT_0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

which allows to express the end-effector pose eventually in the base frame as

$${}^bT_{EE} = {}^bT_0 \cdot {}^0T_{EE} \quad (5)$$

The derivation of individual transformation matrices is omitted here for the sake of simplicity, with expressions of each element of ${}^bT_{EE}$ given as follows:

$$\begin{aligned} t_{11} &= -((c\theta_1 \cdot c\theta_4 \cdot s\theta_{23} + s\theta_1 \cdot s\theta_4) \cdot c\theta_5 + c\theta_1 \cdot s\theta_5 \cdot c\theta_{23}) \cdot c\theta_6 \\ &\quad + (c\theta_1 \cdot s\theta_4 \cdot s\theta_{23} - s\theta_1 \cdot c\theta_4) \cdot s\theta_6 \\ t_{21} &= -((s\theta_1 \cdot c\theta_4 \cdot s\theta_{23} - c\theta_1 \cdot s\theta_4) \cdot c\theta_5 + s\theta_1 \cdot s\theta_5 \cdot c\theta_{23}) \cdot c\theta_6 \\ &\quad + (s\theta_1 \cdot s\theta_4 \cdot s\theta_{23} + c\theta_1 \cdot c\theta_4) \cdot s\theta_6 \\ t_{31} &= (s\theta_5 \cdot s\theta_{23} - c\theta_4 \cdot c\theta_5 \cdot c\theta_{23}) \cdot c\theta_6 + s\theta_4 \cdot s\theta_6 \cdot c\theta_{23} \\ t_{12} &= ((s\theta_1 \cdot s\theta_4 + c\theta_1 \cdot c\theta_4 \cdot s\theta_{23}) \cdot c\theta_5 + c\theta_1 \cdot s\theta_5 \cdot c\theta_{23}) \cdot s\theta_6 \\ &\quad + (s\theta_4 \cdot s\theta_{23} \cdot c\theta_1 - s\theta_1 \cdot c\theta_4) \cdot c\theta_6 \\ t_{22} &= ((s\theta_1 \cdot c\theta_4 \cdot s\theta_{23} - c\theta_1 \cdot s\theta_4) \cdot c\theta_5 + s\theta_1 \cdot s\theta_5 \cdot c\theta_{23}) \cdot s\theta_6 \\ &\quad + (s\theta_1 \cdot s\theta_4 \cdot s\theta_{23} + c\theta_1 \cdot c\theta_4) \cdot c\theta_6 \\ t_{32} &= -(s\theta_5 \cdot s\theta_{23} - c\theta_4 \cdot c\theta_5 \cdot c\theta_{23}) \cdot s\theta_6 + s\theta_4 \cdot c\theta_6 \cdot c\theta_{23} \\ t_{13} &= -(c\theta_1 \cdot c\theta_4 \cdot s\theta_{23} + s\theta_1 \cdot s\theta_4) \cdot s\theta_5 + c\theta_1 \cdot c\theta_5 \cdot c\theta_{23} \\ t_{23} &= -(s\theta_1 \cdot c\theta_4 \cdot s\theta_{23} - c\theta_1 \cdot s\theta_4) \cdot s\theta_5 + s\theta_1 \cdot c\theta_5 \cdot c\theta_{23} \\ t_{33} &= -s\theta_5 \cdot c\theta_4 \cdot c\theta_{23} - s\theta_{23} \cdot c\theta_5 \\ p_x &= a_2 \cdot c\theta_1 + a_3 \cdot s\theta_2 \cdot c\theta_1 + a_4 \cdot s\theta_{23} \cdot c\theta_1 + d_{4'} \cdot c\theta_1 \cdot c\theta_{23} + d_{L'} + q_1 \\ &\quad - d_6 \cdot ((s\theta_1 \cdot s\theta_4 + c\theta_1 \cdot c\theta_4 \cdot s\theta_{23}) \cdot s\theta_5 - c\theta_1 \cdot c\theta_5 \cdot c\theta_{23}) \\ p_y &= a_2 \cdot s\theta_1 + a_3 \cdot s\theta_1 \cdot s\theta_2 + a_4 \cdot s\theta_1 \cdot s\theta_{23} + a_{LF} + d_{4'} \cdot s\theta_1 \cdot c\theta_{23} \\ p_z &= a_3 \cdot c\theta_2 + a_4 \cdot c\theta_{23} - a_{L'} + d_1 - d_6 \cdot (c\theta_4 \cdot s\theta_5 \cdot c\theta_{23} + c\theta_5 \cdot s\theta_{23}) - d_{4'} \cdot s\theta_{23} \end{aligned} \quad (6)$$

Note that $s\theta_i$ and $c\theta_i$ stand for $\sin(\theta_i)$ and $\cos(\theta_i)$, θ_{23} refers to $(\theta_2 + \theta_3)$.

3 Inverse Kinematics

As well-known characteristics for 6-DoF robotic arm, the first three joints, casually referred to as shoulder pan (θ_1), shoulder lift (θ_2) and elbow (θ_3), dominate principally the positioning of the end-effector, while the last three joints (referenced as wrists) determine the finite orientation of the end-effector. The system is under-determined, as the base of the 6-DoF arm is attached to the linear axis, the latter of which also impacts the

translation (i.e., p_x) of the end-effector. A new constraint could therefore be added to solve the system, which consists of constraining θ_1 as follows.

Constraint on θ_1 :

$$\theta_1 = \begin{cases} \frac{\pi}{2}, & \text{if } p_y \geq 0, \\ -\frac{\pi}{2}, & \text{if } p_y < 0. \end{cases} \quad (7)$$

Solution of q_1 :

The solution of q_1 is derived using the position of the 5-th joint of the ABB robotic arm. The transformation from frame 0 to frame 5 is first derived, and the following equality condition can be formed.

$$({}^bT_0)^{-1}({}^bT_{EE}^d)({}^5T_{EE})^{-1} = {}^0T_L(q_1) \cdot {}^LT_{L'} \cdot {}^{L'}T_{LF} \cdot {}^{LF}T_1(\theta_1) \cdot {}^1T_2(\theta_2) \cdot {}^2T_3(\theta_3) \cdot {}^3T_4(\theta_4) \cdot {}^4T_{4'} \cdot {}^{4'}T_5(\theta_5) \quad (8)$$

The z -coordinate of the position in the above equality condition yields

$$p_x - d_6 \cdot t_{13} = a_2 \cdot c\theta_1 + a_3 \cdot s\theta_2 \cdot c\theta_1 + a_4 \cdot s\theta_{23} \cdot c\theta_1 + d_{4'} \cdot c\theta_1 \cdot c\theta_{23} + d_{L'} + q_1 \quad (9)$$

Considering the condition $\cos(\theta_1) = 0$ from (7), the above condition is further simplified to solve q_1 as:

$$q_1 = p_x - d_6 \cdot t_{13} - d_{L'} \quad (10)$$

Solutions of θ_2 and θ_3 :

The solutions of θ_2 and θ_3 are then found by expressing the position of frame 5 relative to frame LF and forming the equality condition as follows

$$({}^bT_{LF})^{-1}({}^bT_{EE}^d)({}^5T_{EE})^{-1} = {}^{LF}T_1(\theta_1) \cdot {}^1T_2(\theta_2) \cdot {}^2T_3(\theta_3) \cdot {}^3T_4(\theta_4) \cdot {}^4T_{4'} \cdot {}^{4'}T_5(\theta_5) \quad (11)$$

The x and y -coordinates of the position of the above equality condition yield

$$\begin{cases} -a_{LF} - d_6 \cdot t_{23} + p_y = (a_2 + a_3 \cdot s\theta_2 + a_4 s\theta_{23} + d_{4'} c\theta_{23}) s\theta_1 \\ a_{L'} - d_6 \cdot t_{33} + p_z = a_3 \cdot c\theta_2 + a_4 c\theta_{23} + d_1 - d_{4'} s\theta_{23} \end{cases} \quad (12)$$

which can be further factorized as

$$\begin{cases} a_3 \cdot s\theta_2 = -a_4 \cdot s\theta_{23} - d_{4'} \cdot c\theta_{23} + z_1 \\ a_3 \cdot c\theta_2 = -a_4 \cdot c\theta_{23} + d_{4'} \cdot s\theta_{23} + z_2 \end{cases} \quad (13)$$

with $z_1 = \frac{p_y - a_{LF} - d_6 \cdot t_{23}}{s\theta_1} - a_2$ and $z_2 = a_{L'} - d_6 \cdot t_{33} + p_z - d_1$.

Equation (13) can be solved using trigonometric relations, which is done firstly by computing the squares of the left-hand and right-hand terms of (13) to eliminate θ_2 and find the solutions for θ_{23} , followed by solving θ_2 and θ_3 sequentially. Computing the squares of (13) on both sides yields

$$2 \cdot (-a_4 \cdot z_1 + d_{4'} \cdot z_2) \cdot s\theta_{23} - 2 \cdot (a_4 \cdot z_2 + d_{4'} \cdot z_1) \cdot c\theta_{23} = a_3^2 - a_4^2 - d_{4'}^2 - z_1^2 - z_2^2 \quad (14)$$

By setting $X = 2 \cdot (-a_4 \cdot z_1 + d_{4'} \cdot z_2)$, $Y = -2 \cdot (a_4 \cdot z_2 + d_{4'} \cdot z_1)$, and $Z = a_3^2 - a_4^2 - d_{4'}^2 - z_1^2 - z_2^2$, the solutions of θ_{23} could be found considering the trigonometric relation

$$\theta_{23} = \text{atan2} \left(\frac{XZ \pm Y\sqrt{X^2 + Y^2 - Z^2}}{X^2 + Y^2}, \frac{YZ \mp X\sqrt{X^2 + Y^2 - Z^2}}{X^2 + Y^2} \right), \quad (15)$$

given that X , Y and Z are not zero, and $X^2 + Y^2 \geq Z^2$. If it is not the case, θ_{23} will be solved differently using

$$\theta_{23} = \begin{cases} \pm \arccos(\frac{Z}{Y}), & \text{if } X = 0, Y \neq 0; \\ \{t, \pi - t\}, \text{ with } t = \arcsin(\frac{Z}{X}), & \text{if } X \neq 0, Y = 0; \\ \{t, \pi + t\}, \text{ with } t = \operatorname{atan2}(-Y, X), & \text{if } Z = 0; \\ \text{No solution}, & \text{otherwise.} \end{cases} \quad (16)$$

By replacing the solutions of θ_{23} in the right-hand side of (13), θ_2 can be solved as

$$\theta_2 = \operatorname{atan2} \left(\frac{-a_4 \cdot c\theta_{23} + d_{4'} \cdot s\theta_{23} + z_2}{a_3}, \frac{-a_4 \cdot s\theta_{23} - d_{4'} \cdot c\theta_{23} + z_1}{a_3} \right) \quad (17)$$

which allows to find the solutions of θ_3 simply by $\theta_3 = \theta_{23} - \theta_2$.

Solutions of θ_4 , θ_5 and θ_6 :

Once θ_1 , θ_2 and θ_3 are solved, the solutions for the last three joints can be found by determining the orientation from frame 3 to 6, with the following equality condition that can be established

$${}^3T_4(\theta_4) \cdot {}^4T_{4'} \cdot {}^{4'}T_5(\theta_5) \cdot {}^5T_6(\theta_6) = ({}^bT_3)^{-1} ({}^bT_{EE}^d) ({}^6T_{EE})^{-1} \quad (18)$$

which yields the following equality conditions for the orientation matrix.

$$\begin{bmatrix} -s\theta_4 \cdot s\theta_6 + c\theta_4 \cdot c\theta_5 \cdot c\theta_6 & -s\theta_4 \cdot c\theta_6 - c\theta_4 \cdot c\theta_5 \cdot s\theta_6 & -c\theta_4 \cdot s\theta_5 \\ s\theta_5 \cdot c\theta_6 & -s\theta_5 \cdot s\theta_6 & c\theta_5 \\ -c\theta_4 \cdot s\theta_6 - s\theta_4 \cdot c\theta_5 \cdot c\theta_6 & -c\theta_4 \cdot c\theta_6 + s\theta_4 \cdot c\theta_5 \cdot s\theta_6 & s\theta_4 \cdot s\theta_5 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad (19)$$

with

$$\begin{cases} r_{11} = -t_{11} \cdot c\theta_1 \cdot s\theta_{23} - t_{21} \cdot s\theta_1 \cdot s\theta_{23} - t_{31} \cdot c\theta_{23} \\ r_{21} = -t_{11} \cdot c\theta_1 \cdot c\theta_{23} - t_{21} \cdot s\theta_1 \cdot c\theta_{23} + t_{31} \cdot s\theta_{23} \\ r_{31} = t_{11} \cdot s\theta_1 - t_{21} \cdot c\theta_1 \\ r_{12} = -t_{12} \cdot c\theta_1 \cdot s\theta_{23} - t_{22} \cdot s\theta_1 \cdot s\theta_{23} - t_{32} \cdot c\theta_{23} \\ r_{22} = -t_{12} \cdot c\theta_1 \cdot c\theta_{23} - t_{22} \cdot s\theta_1 \cdot c\theta_{23} + t_{32} \cdot s\theta_{23} \\ r_{32} = t_{12} \cdot s\theta_1 - t_{22} \cdot c\theta_1 \\ r_{13} = t_{13} \cdot c\theta_1 \cdot s\theta_{23} + t_{23} \cdot s\theta_1 \cdot s\theta_{23} + t_{33} \cdot c\theta_{23} \\ r_{23} = t_{13} \cdot c\theta_1 \cdot c\theta_{23} + t_{23} \cdot s\theta_1 \cdot c\theta_{23} - t_{33} \cdot s\theta_{23} \\ r_{33} = -t_{13} \cdot s\theta_1 + t_{23} \cdot c\theta_1 \end{cases} \quad (20)$$

From the equality condition of $c\theta_5 = r_{23}$, θ_5 can be solved by

$$\theta_5 = \pm \arccos(r_{23}) \quad (21)$$

If $s\theta_5 \neq 0$ (i.e., $r_{23} \neq \pm 1$), θ_4 can be derived considering that

$$\begin{cases} -c\theta_4 \cdot s\theta_5 = r_{13} \\ s\theta_4 \cdot s\theta_5 = r_{33} \end{cases} \quad (22)$$

which yields the solution for θ_4 as

$$\theta_4 = \operatorname{atan2} \left(\frac{r_{33}}{s\theta_5}, \frac{-r_{13}}{s\theta_5} \right) \quad (23)$$

Solving θ_6 similarly using

$$\begin{cases} s\theta_5 \cdot c\theta_6 = r_{21} \\ -s\theta_5 \cdot s\theta_6 = r_{22} \end{cases} \quad (24)$$

yields

$$\theta_6 = \text{atan2}\left(\frac{-r_{22}}{s\theta_5}, \frac{r_{21}}{s\theta_5}\right) \quad (25)$$

If $s\theta_5 = 0$, i.e., $\theta_5 = 0$ must be valid given the angle limits of θ_5 ($|\theta_5| < \pi$), the configuration refers to the wrist singularity condition where joints 4 and 6 become aligned, resulting θ_4 , and θ_6 being under-determined and the robot losing one DoF in orientation. However, $\theta_{46} = (\theta_4 + \theta_6)$ can be solved using

$$\begin{cases} c\theta_4 \cdot c\theta_6 - s\theta_4 \cdot s\theta_6 = c\theta_{46} = r_{11} \\ -(c\theta_4 \cdot s\theta_6 + s\theta_4 \cdot c\theta_6) = -s\theta_{46} = r_{31} \end{cases} \quad (26)$$

as

$$\theta_{46} = \text{atan2}(-r_{31}, r_{11}) \quad (27)$$

In such cases, to eventually determine θ_4 and θ_6 , their values can be set to

$$\begin{cases} \theta_4 = 0 \\ \theta_6 = \text{atan2}(-r_{31}, r_{11}) \end{cases} \quad \text{or} \quad \begin{cases} \theta_4 = \text{atan2}(-r_{31}, r_{11}) \\ \theta_6 = 0 \end{cases} \quad (28)$$

After all the candidate sets of joint positions are found, the best candidate is selected such that the transition from the current joint configuration to the desired one is minimum, i.e., the shortest distance between the desired joint position and the current one in the joint configuration space is selected. The distance is computed by finding the joint position differences component-wise and computing the norm as

$$\begin{aligned} dist^j &= \text{norm}(|\mathbf{q}^j - \mathbf{q}_{curr}|) \\ |\mathbf{q}^j - \mathbf{q}_{curr}| &= \begin{cases} |q_1^j - q_{1,curr}| \\ \text{wrap}(\theta_i^j - \theta_{i,curr}), \quad \text{for } i \in [1, 6] \end{cases} \end{aligned} \quad (29)$$

where $\mathbf{q}^j$ is one candidate set of joint positions, $\mathbf{q}_{curr}$ is the current joint configuration, $\text{norm}(\cdot)$ represents the 2-norm distance of a vector, and $\text{wrap}(\cdot)$ is a function that wrap angles between $[0, 2\pi]$.

4 Simulation Validation

In order to validate the IK solutions found in the above section, real-time simulation environment is set up using ROS2 and Gazebo. In this very first setup of the simulator, there is a window dialog allowing user to input a desired end-effector pose by determining its position and Euler angles (in roll-pitch-yaw order) as orientation. The IK solver then computes the corresponding joint positions to allow the end-effector to reach that pose. A few examples of solving joint positions from a given desired end-effector pose, along with visualization in RViz (shown in Figures 2 and 3), are demonstrated below.

Example 1:

Desired end-effector pose:

$$p_x = 1.5, p_y = 3.2, p_z = 1.78, \text{roll} = -90^\circ, \text{pitch} = 90^\circ, \text{yaw} = 0^\circ.$$

Joint solutions:

$$q_1 = 0.9, \theta_1 = 1.571, \theta_2 = 0.631, \theta_3 = -0.134, \theta_4 = -3.142, \theta_5 = 0.497, \theta_6 = 3.142.$$

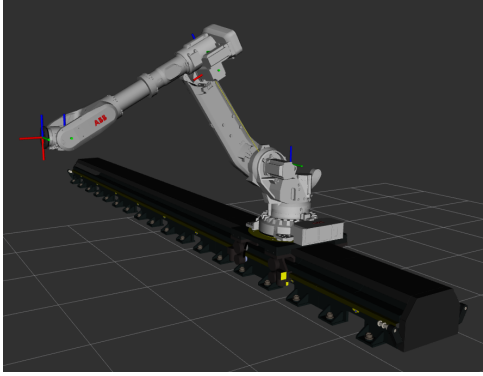
Example 2:

Desired end-effector pose:

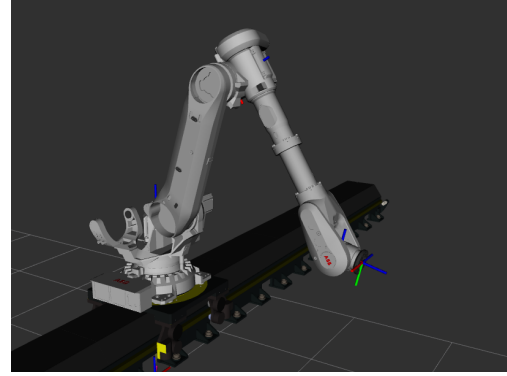
$$p_x = 3.6, p_y = -1.6, p_z = 1.2, \text{roll} = 80^\circ, \text{pitch} = 180^\circ, \text{yaw} = 0^\circ.$$

Joint solutions:

$$q_1 = 3.0, \theta_1 = -1.571, \theta_2 = 0.421, \theta_3 = 0.610, \theta_4 = 0, \theta_5 = -0.804, \theta_6 = -1.571.$$



(a) Example 1



(b) Example 2

Figure 2: Robot configuration in two examples.

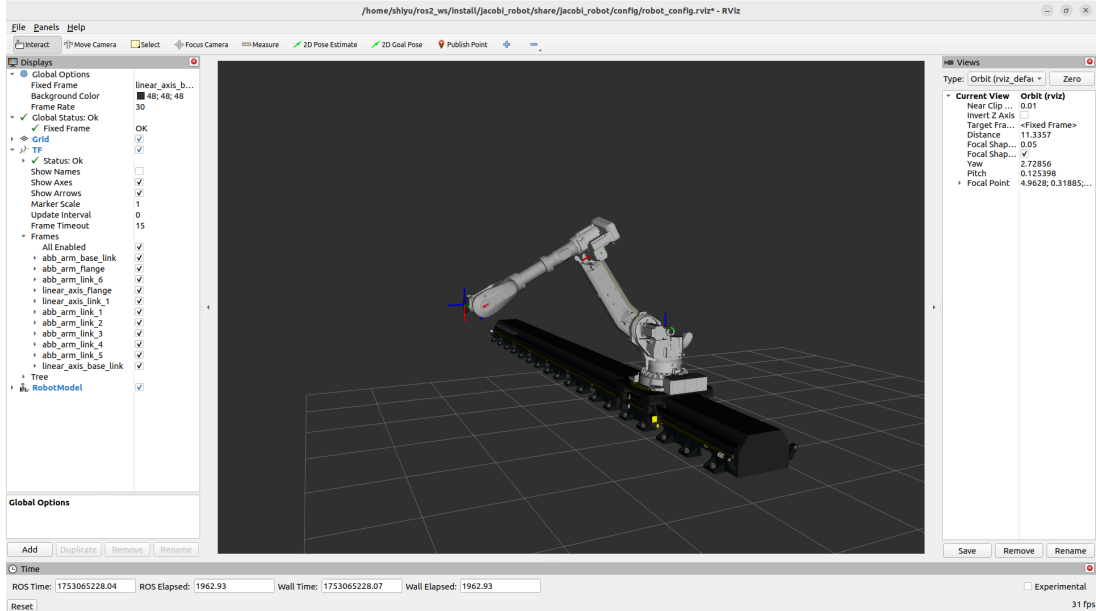


Figure 3: RViz visualization